



Immunology of IBD & Celiac Disease — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions**1. Which antibody is more commonly raised in ulcerative colitis?**

EASY

- A. ASCA
- B. Anti-tTG
- C. Anti-mitochondrial
- D. ANA
- E. pANCA

2. Which HLA alleles are strongly associated with celiac disease?

EASY

- A. HLA-DQ2/DQ8
- B. HLA-B27
- C. HLA-DR*103 only
- D. HLA-DR3/DR4
- E. HLA-A2

3. What is the treatment for celiac disease?

EASY

- A. Anti-TNF therapy
- B. Lactose avoidance
- C. Corticosteroids lifelong
- D. Lifelong gluten avoidance
- E. Surgical resection

4. Which IBD is characterised by skip lesions?

EASY

- A. Ulcerative colitis
- B. Crohn's disease
- C. Both equally
- D. Celiac disease
- E. Neither

5. A patient with UC stopped smoking and then had a flare. Which statement fits the lecture?

MEDIUM

- A. Smoking is protective for UC but a risk factor for CD
- B. Smoking causes UC
- C. Smoking is protective for CD
- D. Smoking has no effect on either
- E. Smoking causes celiac disease

6. A colon biopsy shows granulomas with focal inflammation and stricturing. Which disease?

MEDIUM

- A. Ulcerative colitis
- B. Celiac disease
- C. Ischemic colitis
- D. Pseudomembranous colitis
- E. Crohn's disease

7. A mutation in *NOD2/CARD15* increases susceptibility to which disease?

MEDIUM

- A. Crohn's disease
- B. Ulcerative colitis
- C. Celiac disease
- D. Food allergy
- E. Lactose intolerance

8. Which T-cell change characterises IBD?

MEDIUM

- A. ↓ Th17 and ↑ Treg cells
- B. ↑ Th17 and ↓ Treg cells
- C. ↑ Treg only
- D. ↑ Th2 only
- E. No change

9. A patient has diarrhea and abdominal pain; which test separates active from inactive IBD?

MEDIUM

- A. Anti-tTG IgA
- B. Serum ferritin
- C. ANA
- D. Fecal calprotectin
- E. Stool culture

10. A patient has pseudopolyps and severe colitis with a thinned bowel wall. Which disease and complication?

MEDIUM

- A. Crohn's with granuloma
- B. Celiac with atrophy
- C. Ulcerative colitis with toxic megacolon
- D. Ischemic bowel
- E. IBS

11. A patient with IBD has joint inflammation without rheumatoid factor and uveitis. Which additional extra-intestinal feature is listed in the lecture?

MEDIUM

- A. Erythema chronicum migrans
- B. Optic neuritis due to demyelination
- C. Pancreatic calcification
- D. Hyperthyroid goiter
- E. Sclerosing cholangitis

12. Which cereals contain gluten-type proteins that trigger celiac disease?

MEDIUM

- A. Rice and maize
- B. Wheat, rye, barley (less oats)
- C. Oats only
- D. Soy and rice
- E. Quinoa and millet

13. What is the gold standard for celiac diagnosis?

MEDIUM

- A. Stool culture
- B. Colonoscopy with ileal biopsy
- C. Lactulose breath test
- D. Duodenal biopsy combined with positive serology
- E. Skin-prick test

14. A woman presents with anemia, osteopenia and infertility but no diarrhea. Which form of celiac disease?

MEDIUM

- A. Classic celiac disease
- B. Silent celiac disease
- C. Atypical celiac disease
- D. Whipple disease
- E. Tropical sprue

15. Anti-tTG IgA is negative in a patient with strongly suggestive villous atrophy. Which explanation and action fits?

HARD

- A. The disease is excluded
- B. Repeat with pANCA
- C. Test for ASCA
- D. Total IgA deficiency; measure total IgA and test IgG antibodies
- E. Start anti-TNF

16. Gluten peptides are deamidated in the lamina propria. What is the immunological consequence?

HARD

- A. Enhanced IgE production
- B. Suppression of Th1
- C. Enhanced binding to HLA-DQ2 and IFN- γ Th1 activation
- D. Loss of tTG
- E. Increased Treg

17. A patient with long-standing celiac disease becomes unwell with weight loss and abdominal pain. Which complication is most feared?

HARD

- A. Enteropathy-associated T-cell lymphoma
- B. Colorectal cancer only
- C. Toxic megacolon
- D. Hepatitis
- E. Cholera

18. A patient with IBD has low levels of butyrate-producing bacteria. Which dietary pattern is linked to this?

HARD

- A. Mediterranean diet
- B. High fat and sugar with low fibre
- C. High polyphenols
- D. High fibre
- E. Vegetarian

19. Which combination is expected in severe Crohn's disease?

HARD

- A. Normal albumin and Hb
- B. Hypoglycemia only
- C. Thrombocytopenia only
- D. Hyponatremia only
- E. Hypoalbuminemia, anemia and leukocytosis

20. Corticosteroids and 5-ASA compounds treat IBD by:

HARD

- A. Blocking $\alpha 4\beta 7$ integrin
- B. Neutralising TNF directly
- C. Inhibiting NF- κ B-regulated inflammatory mediators
- D. Replacing lost microbiota
- E. Depleting B cells

✓ Answer key

Q	Answer	Difficulty	Why
1	E	EASY	pANCA is more common in UC; ASCA in CD.
2	A	EASY	Celiac disease has a strong HLA-DQ2/DQ8 association.
3	D	EASY	The bowel normalises when gluten is withdrawn.
4	B	EASY	CD can affect any part of the GIT with skip lesions.
5	A	MEDIUM	UC is more common in non- and ex-smokers.
6	E	MEDIUM	Granulomas, fistulas and strictures are typical of CD.
7	A	MEDIUM	NOD2 senses muramyl dipeptide; mutations strongly link to CD.
8	B	MEDIUM	Th17 drives inflammation; the IL-23/Th17 pathway is central.
9	D	MEDIUM	Calprotectin differentiates active from inactive disease and monitors it.
10	C	MEDIUM	Pseudopolyps and toxic megacolon are UC findings.
11	E	MEDIUM	Seronegative arthritis, uveitis and sclerosing cholangitis are the listed extra-intestinal manifestations.
12	B	MEDIUM	Gliadin and similar proteins occur in wheat, rye, barley and to a lesser extent oats.
13	D	MEDIUM	Mucosal changes plus tTG / EmA / DGP serology.
14	C	MEDIUM	Atypical disease has non-intestinal manifestations.
15	D	HARD	Low IgA anti-tTG can result from total IgA deficiency.
16	C	HARD	tTG deamidates glutamine residues, creating a T-cell epitope that binds DQ2.
17	A	HARD	Celiac disease carries about twofold increased malignancy risk, particularly this lymphoma.
18	B	HARD	Such diets reduce SCFA producers and raise pathogens.
19	E	HARD	Severe disease shows hypoalbuminemia, anemia and leukocytosis with elevated ESR/CRP.
20	C	HARD	Anti- $\alpha 4\beta 7$ is an integrin antagonist; steroids and 5-ASA act via NF- κ B.